EvrLight: Asset Payments on Lightning

hans_schm1dt@protonmail.com 2026-10-27



Beyond Zaps and Value-4-Value:
Use Bitcoin Lightning or Block Inc's CashApp to buy and deliver tickets, coupons, NFTs, and real-world tokenized real estate, commodities and securities.

TLDR: EvrLight is an open protocol which uses Nostr for communications and assets residing on a UTXO-based blockchain to enable cross-chain submarine swaps between Bitcoin Lightning payments and assets. EvrLight provides a model for Bitccoin to achieve large-scale adoption as the medium of exchange for the sale of assets without burdening the chain with ordinals, Runes, BRC-20, or other similar schemes.

Bitcoin is the world's best type of cash money and is now widely used as a store of value. But Bitcoin cannot succeed long term if it does not also become a mainstream medium of exchange. The Lightning network helps by enabling very fast transactions at very low fee cost. The invention of Zaps on Nostr has showcased Bitcoin Lightning for peer-to-peer cash. Bolt12, Lightning service providers, and LDK are further improving the Lightning user experience. But we must do more.

EvrLight: Asset Payments on Lightning

hans_schm1dt@protonmail.com 2026-10-27

Cash is only one of many forms of value which are exchanged in daily commerce. There are many forms of natively digital assets such as tickets, coupons and loyalty points. Even NFTs, which have dubious value when simply representing image files, have great utility as trade-able serialized event tickets with assigned seating, for example. Additionally, there is a growing number of efforts globally to tokenize real estate, commodities and securities. Combined, these asset types have value far larger than the amount of cash in the world.

But Bitcoin is not benefiting from the asset marketplaces (with the exception of some stablecoins). There are good reasons for this. Any effort to represent an asset on the Bitcoin blockchain, regardless of how it is implemented, inevitably increases the chain size by adding extra data to the chain, which some people (rightfully or wrongly) argue is bad for Bitcoin. Additionally, trading those assets requires L2 support to achieve speed and low fees. For that purpose, Payment Channels (Lightning) fail entirely. Asset functionality on Lightning has been claimed by some projects such as Taproot Assets, Spark, and LRC-20, but those only function for high-volume fungible assets such as stablecoins and certainly not for NFTs. Bitcoin provides trillion-dollar security for a better global cash, but doesn't want the burden of data for assets.

The organized markets for asset trading are served by specialized blockchain projects integrated with the legacy financial infrastructure, as well as by a patchwork of smaller blockchains and exchanges, both centralized and distributed, and optimized for asset functional and security needs. For peer-to-peer commerce over social media like Nostr, there is no open source protocol or set of standards for trading assets, and therefore there is no bitcoin-based social marketplace activity.

To increase Bitcoin's use as a medium of exchange and to achieve wide-scale adoption of Lightning, open source protocols and standards are needed which support Bitcoin for asset payments without burdening the chain with ordinals, Runes, BRC-20, or other similar schemes.

EvrLight is a proof-of-concept demo for serving the asset markets by using Bitcoin Lightning for peer-to-peer trustless permission-less atomic social media based commerce.

EvrLight is like WooCommerce for buying any digital asset paid for and delivered using Bitcoin Lightning or using Dollars via Block Inc's CashApp.

EvrLight re-imagines remote wallet functionality to include both Nostr NIP-46 event signing and Nostr Wallet Connect (NWC) Lightning calls. But it goes even further with API calls to support PSBT signing, UTXO discovery and allocation, and refund address inquiry. These make Nostr applications capable of executing atomic cross-chain submarine swaps. As always, Bitcoin serves as Lightning's Layer-1 for money. But asset functionality is supported by Evrmore, a UTXO-based blockchain dedicated to assets. Use of the UTXO architecture eliminates the unbounded security risks of EVM contract-based chains and is ideal for integration with Bitcoin. The Evrmore blockchain was launched in Oct 2022 with no pre-mine and no ICO/IPO. It is based on Bitcoin's code, uses UTXOs and Proof-of-Work, and is fully open-source. Development has been funded entirely by grants from enthusiasts.

EvrLight: Asset Payments on Lightning

hans_schm1dt@protonmail.com 2026-10-27

Evrmore is not a shit coin - unless manure is what you want to tokenize ;-)

EvrLight defines standardized functions and Nostr kinds which make it easy for apps to support flexible social commerce of any type of asset. EvrLight also supports generating an internet URL which can be dropped into a post on any social media platform, which links directly into a Lightning-based asset purchase dialogue in which the involved parties communicate over Nostr relays. The Evrlight demo is primarily a back-end technology which can optionally also handle the buyer dialogue. For low-value assets, the sale can be considered final as soon as the HTLC cross-chain submarine swap transaction appears in the Evrmore blockchain mempool, just a few seconds after the buyer pays the Lightning invoice. For higher value assets, the first confirmation of the HTLC contract and sweep transactions complete on average about 2 minutes after payment of the Lightning invoice. Faster performance is possible.

The EvrLight demo is not a traditional Nostr client, but it uses Nostr throughout its design. It defines Nostr kinds for sell offers and purchase status updates, and defines a protocol for buyers and sellers to find each other over Nostr relays.

Social media based commerce today exists only within corporate walled gardens, using traditional fiat payments rails controlled by the owner. For global permission-less social commerce, Bitcoin would have advantage over fiat-based transactions constrained by jurisdictional regulations. The EvrLight protocol provides the protocol and standards to support peer-to-peer trustless permission-less atomic sale of assets paid for and delivered by Lightning on Nostr and other social media. Lightning-based social commerce would help support small businesses and entrepreneurs. EvrLight is for small town Main street merchants and self-employed entrepreneurs, not only New York Wall street flash boys. Nostr has shown that it can provide the protocol for our on-line lives in which we build webs-of-trust with the people with whom we choose to build relationships, without the needless risk of trusting unnecessary middlemen or soulless corporations. Likewise in commerce, EvrLight aims to enable a thriving economy of small business tokenization experts and integrators providing the technology and services to support peer-to-peer social commerce for their non-technical clients, proving their trustworthiness for inclusion in the webs-of-trust of their customers while themselves choosing who to engage with. EvrLight social commerce is about young vocalists who want to create and sell tokens to their fans so that they have some income while working on their next album, while their fans can trade or use their tokens to get pre-release copies of the new album when it is ready. EvrLight social commerce is about small churches who can organize fund-raising by providing their parishioners with tokens to sell or use at full price for purchases at local merchants with whom the church negotiated discounts as donations.

Very few assets today exist on the Bitcoin blockchain, which EvrLight will not change. Hopefully, EvrLight will encourage asset marketplaces to use Evrmore or to start an Evrmore-like blockchain on which their assets will exist. The EvrLight protocol can then be used for all asset sales, thereby using Lightning to conduct payments in Bitcoin. The participants will be saved from the inefficiencies of barter and from the risks of needing to use stablecoins. Bitcoin will benefit from large-scale adoption as the medium of exchange.

EvrLight: Asset Payments on Lightning

hans_schm1dt@protonmail.com 2026-10-27

Current status:

- EvrLight is currently running as a proof-of-concept demo on Bitcoin testnet-3 and Evrmore testnet
- Additional testing and debugging needed
- More code documentation needed
- Documentation of protocol needed for standardization
- Needs work making LND Lightning node setup easier

Future direction:

- A customized Nostr app with a seamless EvrLight UI
- Fund and co-develop some real-world specialized use case websites/apps
- A distributed buyer/seller/swapservice model in which commissions are paid to both merchant and buyer gateway communities

For more information:

- Demo screen shots: https://hans-schmidt.github.io/mastering_evrmore/evrlight_and_nostr4evr/demo/evrlight_demo_walkthrough.html

- UTXO asset example: https://hans-schmidt.github.io/mastering_evrmore/current_tech_docs/bitcoin_vs_evrmore_txn_comparison.html

- EvrLight info: https://hans-schmidt.github.io/mastering_evrmore/evrlight_and_nostr4evr/evrlight_and_nostr4evr.html

- Evrmore: https://evrmore.com

- hans_schm1dt@protonmail.com